

Photonics in AI Infrastructure: from weak signal to governed factor

How ARKA compiled a freeform photonics thesis into a quarterly rebalanced, auditable long-only factor spanning optical interconnects, transceivers, silicon photonics, packaging, and deployment infrastructure.

Executive view

Photonics first appeared as a fragmented market narrative rather than as a clean sector object. ARKA’s advantage was to detect the theme across standards, platform announcements, component bottlenecks, packaging, and onshoring — then compile that narrative into an explicit exposure model rather than wait for consensus.

What the platform actually did

The engine did not stop at theme discovery. It auto-generated a mathematical framework, translated that framework into deterministic score and weight rules, orchestrated the build across filings, transcripts, patents, standards, and market data, and shipped a portfolio-ready output with an audit trail.

Current ARKA materials

1.54 Sharpe

136.78% ann. return
2024-01 to 2026-03

Archive post-trigger

2.71 Sharpe

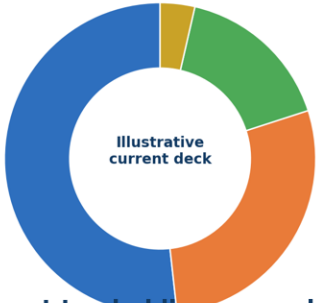
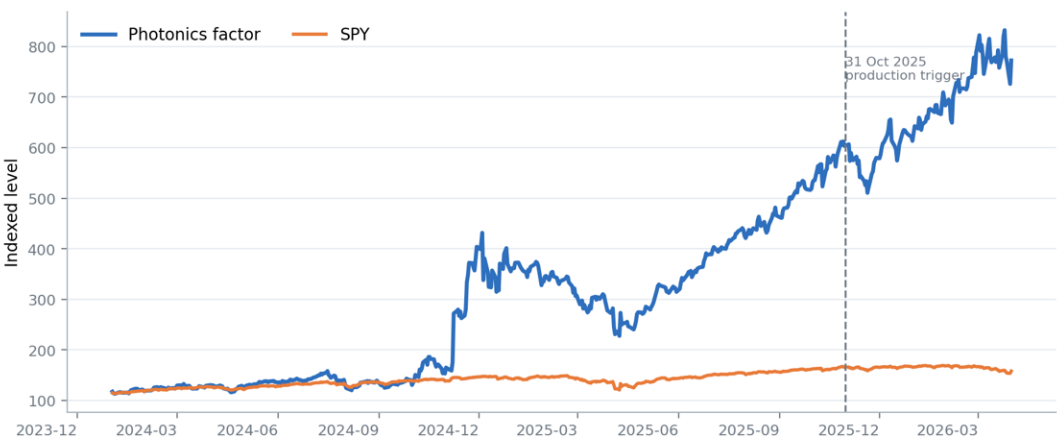
66.4% total return
31 Oct 2025 to 13 Apr 2026

Archive full history

903.9% total

51.6% alpha
31 Mar 2021 to 13 Apr 2026

Cumulative return since Jan 2024 (indexed to 100)



Current sub-industry mix

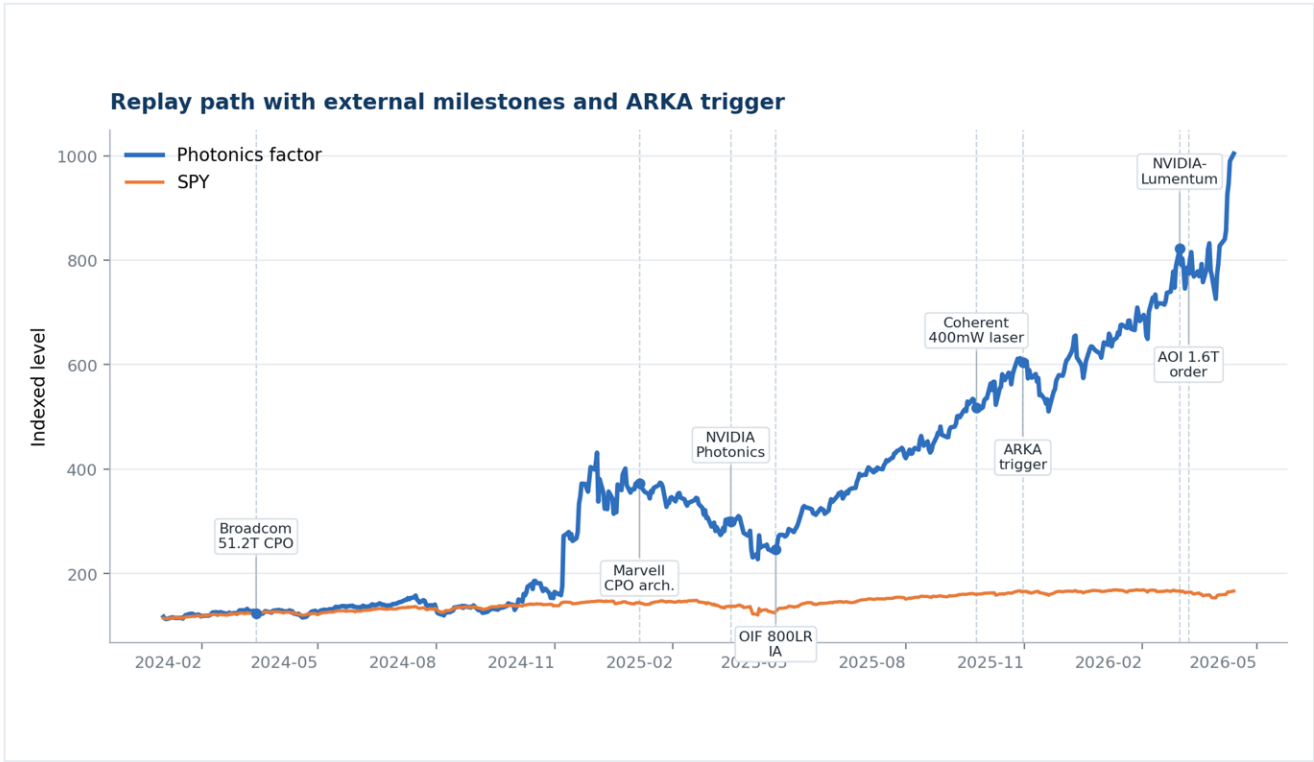
- Semiconductors 51.7%
- Semi. Materials & Equip. 28.2%
- Electronic Components 16.4%
- Communications Equip. 3.6%

Current top holdings snapshot

Ticker	Weight	Industry
AAOI	11.72%	Tech. Hardware
GLW	11.34%	Tech. Hardware
MXL	10.22%	Semiconductor
NVDA	9.99%	Semiconductor
MTSI	9.28%	Semiconductor

How the theme was identified — and when the signal broke out

The edge was not that photonics was unknown. It was that the signal emerged across multiple channels before it existed as a clean portfolio object: standards, platform architecture, component bottlenecks, packaging, and domestic buildout.



What ARKA was seeing

- Standards hardening: OIF formalized 3.2T CPO and later 800ZR / 800LR modules, clarifying the interoperability path from pluggables to integrated optics.
- Platform architecture: Broadcom’s Bailly CPO switch and NVIDIA Spectrum-X / Quantum-X Photonics made optics a system-level AI fabric issue.
- Component readiness: Coherent, Lumentum, and AOI were highlighting higher-power CW / ELS solutions for SiPh and CPO architectures.
- Module and DSP ramp: Marvell, MaxLinear, and AOI were validating 1.6T / 200G-per-lane readiness for scale-up and scale-out AI networks.
- Packaging and onshoring: Amkor and Coming showed the buildout extending into assembly, fiber, and domestic capacity.

Why this matters

In ARKA’s framework, early detection is not a single headline. It is the convergence of evidence density, technical validation, and price confirmation across the stack.

Internal trigger and run

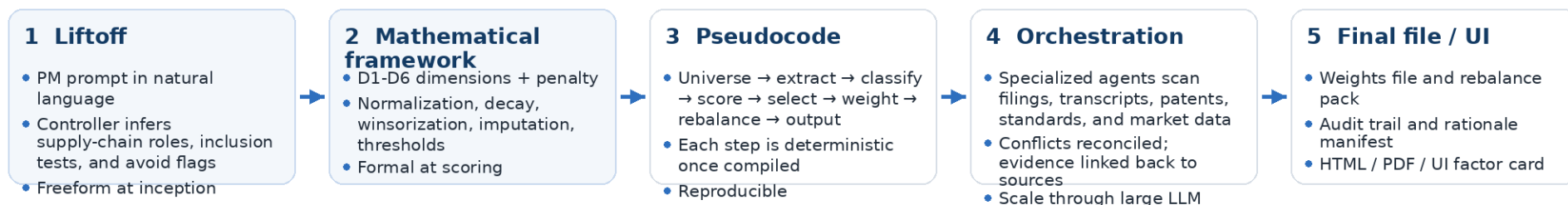
User-supplied chronology: PM prompt submitted 31 Oct 2025; controller launched a multi-hour run on the U.S. equity universe, fusing filings, patents, transcripts, standards, and market data into a quarterly rebalanced long-only factor.

Selected milestones used to frame the signal

Date	Source	Why it mattered
14 Mar 2024	Broadcom	First 51.2T CPO Ethernet switch for scalable AI systems
18 Mar 2025	NVIDIA	Spectrum-X / Quantum-X Photonics; 1.6T per port, 3.5x energy savings claim
22 Apr 2025	OIF	800LR IA released for low-power 800G coherent links up to 10 km
25 Sep 2025	Coherent	400 mW CW lasers sampled for CPO and silicon photonics
31 Oct 2025	ARKA	Production trigger; prompt compiled into factor run
2-9 Mar 2026	Lumentum / AOI	NVIDIA-Lumentum agreement; AOI 1.6T volume order

Controller workflow: from prompt to portfolio-ready output

The attached framework shows two layers working together. The first is freeform and thesis-aware: ARKA infers the right ontology and evidence stack. The second is explicit and governed: once compiled, the factor follows deterministic score, weighting, and rebalance rules.



Freeform at inception → formal at scoring → deterministic at portfolio → auditable at output

Compiled exposure dimensions from the attached framework

Dimension	What it is capturing
D1 Revenue exposure	Explicit SiPh / CPO revenue; product mentions
D2 Speed-node & wins	800G / 1.6T / 3.2T, sample / qualification, hyperscaler wins
D3 Critical components	Laser / modulator IP, vertical stages, component capacity
D4 Packaging capacity	SiPh foundry, 2.5D / 3D / CPO assembly evidence
D5 CPO momentum	Roadmap presence + win / qualification count
D6 Domestic advantage	US footprint, CHIPS / onshoring / long-term supply
Penalty	Pure-pluggable focus or concentrated China-material risk

Core portfolio equations

$$\text{Score}_i = (0.35 \cdot D1 + 0.25 \cdot D2 + 0.20 \cdot D3 + 0.10 \cdot D4 + 0.05 \cdot D5 + 0.05 \cdot D6) \times \text{Penalty}_i$$

$$w_i = \min(0.08, \max(0.005, S_i^2 / \sum S^2));$$
 then apply industry and liquidity caps
 Quarterly rebalance, monthly tradability checks, turnover guard, and selection buffers

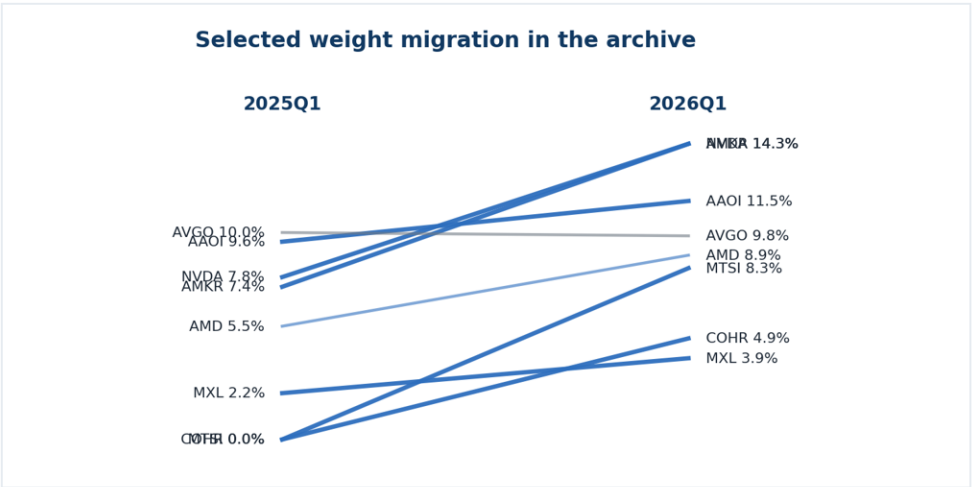
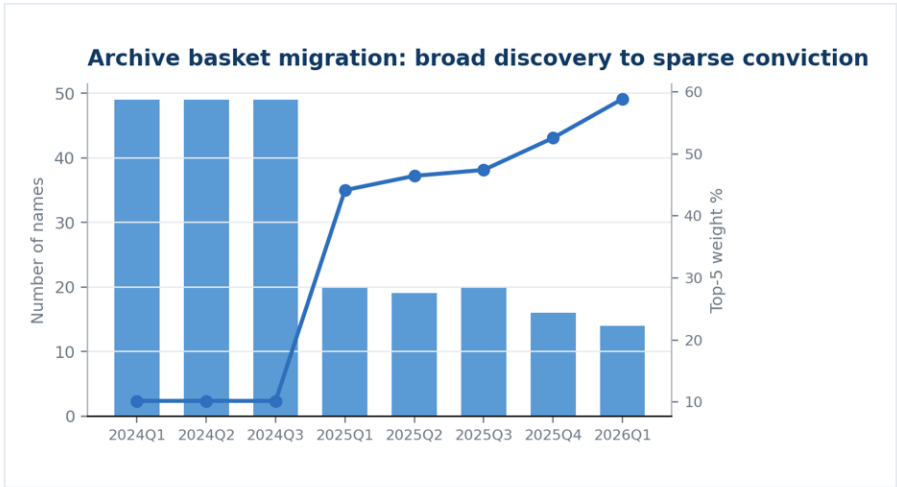
Controller pseudocode

```

Universe <- US-listed semis, optics, packaging, networking adjacencies
Filter <- cap, ADV, price, listing, tradability thresholds
Evidence <- filings + transcripts + patents + standards + price validation
Classify <- revenue, design wins, speed nodes, capacity, onshoring, avoid flags
Score <- normalize -> winsorize -> aggregate dimensions -> apply penalty
Select <- rank, buffer, exclude avoid-list names
Weight <- S^2 subject to single-name, industry, and liquidity caps
Maintain <- quarterly rebalance + monthly tradability maintenance
Output <- weights file + audit trail + report + UI card
At production scale the controller coordinates many specialized agent passes and large-scale LLM calls.
  
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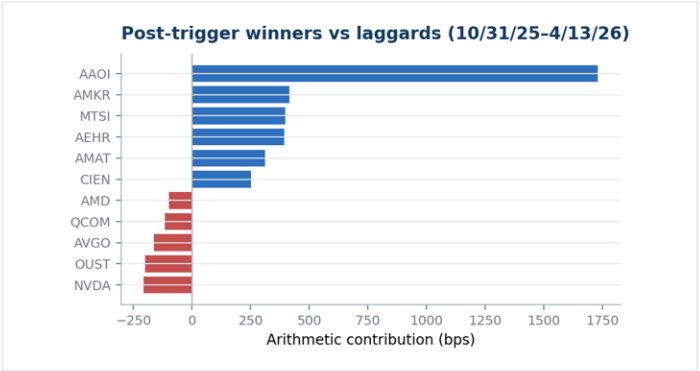
Basket construction: explainable exposures, not theme drift

Archive-based rebalance history shows a shift from a 49-name discovery basket in 2024 to a much sparser expression by 2026Q1. The key point is not concentration by itself; it is that the names gaining weight are explainable by stack position, design-win evidence, component bottlenecks, packaging readiness, and liquidity governance.



Representative supply-chain map

Stack layer	Representative names	Role in the basket
System / switch platform	NVDA, AVGO, AMD, CSCO	Where optical adoption is monetized in the fabric: switch silicon, GPU adjacency, and networking platforms with direct pull-through into optical density.
Modules / connectivity	AAOI, GLW, CIEN, LITE	Optical engines, transceivers, fiber, and interconnect density for data-center scale-up and scale-out links.
Lasers / DSP / analog	MTSI, MXL, COHR	The laser, driver, TIA, and DSP layer that makes higher lane rates and coherent / CPO architectures practical.
Packaging / assembly	AMKR, AMAT	SiPh / CPO packaging, advanced assembly, and manufacturing leverage once the architecture moves from design to volume.



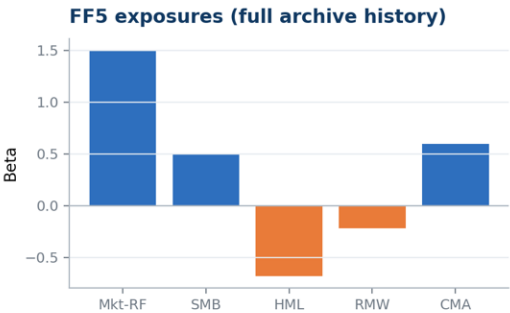
Interpretation
Post-trigger archive contribution was carried by specialists such as AAOI, AMKR, MTSI and COHR. The positive side was therefore not just megacap AI beta; it moved down the stack into optical bottlenecks, DSP / analog, and packaging readiness.

Performance, decomposition, and distinctiveness

Headline returns are strongest in the current ARKA deck window, but PM diligence depends on what survives style controls and concentration tests. The archive suggests a differentiated factor with meaningful residual alpha — not a market-neutral one, but also not fully explained by conventional equity styles.

Three performance lenses

Window	Sharpe	Ann. ret	Ann. vol	Max DD	Note
Current ARKA materials 2024-01 to 2026-03	1.54	136.78%	66.2%	-49.1%	Latest deck KPI set
Archive post-trigger 31 Oct 2025 to 13 Apr 2026	2.71	217.6%	46.7%	-15.8%	Archive validation window
Archive full history 31 Mar 2021 to 13 Apr 2026	1.14	58.0%	50.7%	-48.4%	Archive full back history

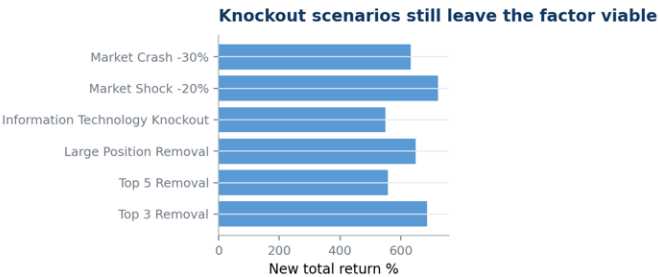


FF5A read-through

Mkt-RF remains the dominant style exposure ($\beta=1.50$), with smaller-cap and growth tilts also visible. That said, model fit is only $R^2=0.373$; residual volatility is 40.1%, and annualized alpha is 51.6% ($t=2.84$, $p=0.0045$).

Institutional wording that is more precise

The right claim is not 'orthogonal.' It is that the factor is distinctive and not fully style-explained after controlling for common equity factors.



Distinctiveness

SPY corr 0.584

SPY beta 1.736

FF5 $R^2=0.373$

Concentration

HHI 0.0957 Effective N 10.5 Top-3 weight

40.1%

Alpha analysis

Alpha 51.6%

t-stat 2.84

p-value 0.0045

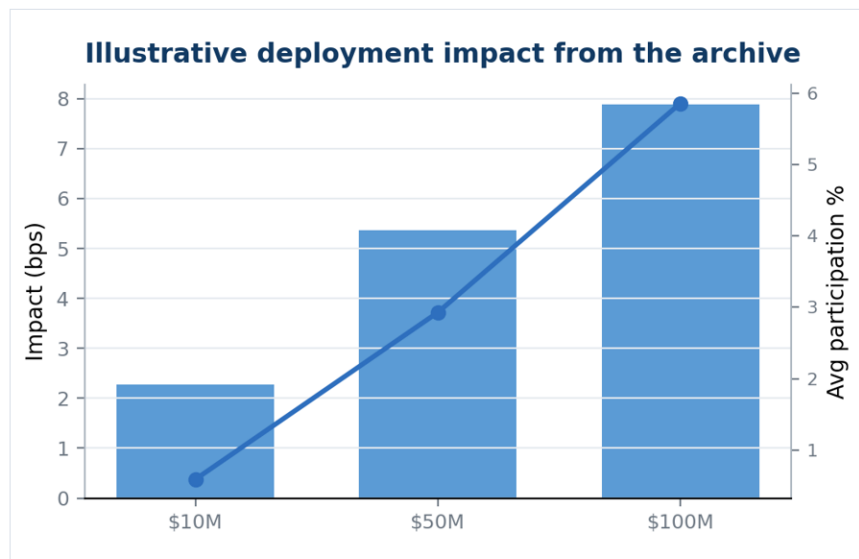
Robustness

Top 3 removal: 686.4% total Top 5 removal: 558.2%

total All six archive scenarios remain viable

Implementation and what this says about the platform

The last test is not thematic elegance but whether the basket survives implementation review. On the archive's own liquidity model it does: the basket is tagged LARGE, with ~\$100m deployment at ~5.9% average participation and ~7.9 bps impact.



Deployability snapshot

Metric	Archive view
Capacity tier	LARGE
Estimated capacity	2422.1 archive units
Avg daily volume	4494.9
Limiting factor	Market impact
Market impact	5.36 bps
Spread cost	2.50 bps
Total cost	8.21 bps

Most capacity-sensitive current archive names

Ticker	Weight	ADV (USD mm)	% ADV for \$100m
AEHR	5.93%	15.9	37.2%
MXL	3.92%	19.7	19.9%
AMKR	14.29%	132.2	10.8%
AAOI	11.52%	156.7	7.4%

1 Not a static theme library

The prompt is freeform. ARKA first generates the relevant ontology, evidence stack, inclusion logic, and avoid flags for the thesis at hand.

2 Formal where it matters

Once the theme is compiled, the factor is constrained by explicit math: normalization, weighting, caps, buffers, turnover guards, and liquidity rules.

3 Self-adapting rather than fixed

As evidence migrates from megacap incumbents to niche bottlenecks, the basket can reallocate accordingly. That is different from a static ETF-like label.

4 Output is portfolio-ready

The endpoint is not an interesting narrative. It is a weights file, a rebalance pack, a rationale manifest, and a UI/report artifact that PMs can review.

Freeform at inception. Disciplined in implementation. Repeatable across emergent supply-chain, geopolitical, and technology shocks.

Selected source notes and chronology

This draft combines internal ARKA materials with official external sources to substantiate the photonics thesis, the controller workflow, and the supply-chain evidence used throughout the case study.

Internal source pack used for this draft

- 1. ARKA platform description and current deck snapshot supplied by the user (headline KPI window, current top holdings, illustrative allocation).
- 2. Photonic factor mathematical framework JSON supplied by the user (D1-D6 dimensions, penalty term, weighting, rebalance, agent workflow, pseudocode).
- 3. Photonics analytics archive supplied by the user (performance history, benchmark comparison, FF5 exposures, rolling metrics, robustness, liquidity, holdings).
- 4. Communication reference only: Morgan Stanley QDS End of Week Wrap, used to calibrate note structure and sell-side cadence rather than factual content.

Chronology note

Chronology note: the production trigger date supplied by the user is 31 Oct 2025. Earlier charts based on the attached analytics archive are shown as historical replay of the compiled methodology. Page 1 deliberately separates the current deck KPI window from archive-based validation windows so the report does not blur live timing with replay history.

Reading guidance

Caveat: where the report cites archive analytics, those results are model-based and should be read as validation exhibits rather than guarantees. Where the report cites external sources, preference was given to official company, standards-body, and primary research publications.

Official external sources referenced in the note

- 1. NVIDIA (18 Mar 2025): Spectrum-X / Quantum-X Photonics announcement; 1.6T-per-port photonics switches and ecosystem collaborators.
- 2. Broadcom (14 Mar 2024): Bailly 51.2T CPO switch press release; first 51.2T CPO Ethernet switch for scalable AI systems.
- 3. OIF (29 Mar 2023; 22 Apr 2025): 3.2T CPO module IA and 800LR IA, framing the standards / interoperability timeline.
- 4. Microsoft Research (Sep 2025): MOSAIC paper/blog describing the copper-versus-optics trade-off in future data-center links.
- 5. Marvell (31 Mar 2025): 1.6T silicon photonics light engine for low-power rack-scale interconnect in AI networks.
- 6. Applied Optoelectronics (28 Feb 2025; 9 Mar 2026): vertically integrated optical manufacturing and first 1.6T hyperscale volume order.
- 7. MaxLinear (31 Mar 2025): Rushmore 1.6T PAM4 DSP for 1.6T optics and active copper cables.
- 8. Coherent (25 Sep 2025; 17 Mar 2026): 400 mW CW lasers plus broad CPO technology stack demonstrations.
- 9. Corning (13 May 2025; 27 Jan 2026): optical components for Broadcom Bailly and multiyear U.S. AI data-center fiber buildout with Meta.
- 10. Lumentum (2 Mar 2026): strategic optics agreement with NVIDIA to develop silicon photonics for AI factories.
- 11. Amkor (20 Feb 2026; 14 Nov 2024): SiPh / CPO packaging capabilities and Lightmatter 3D photonics package partnership.